have selfish or altruistic motives in something like the subjective sense we would recognize. A book called *The Selfish Lion* might actually confuse, in a way that *The Selfish Gene* should not. Just as one can put oneself in the position of an imaginary light beam, intelligently choosing the optimal route through a cascade of lenses and prisms, or an imaginary gene choosing an optimal route through the generations, so one can postulate an individual lioness, calculating an optimal behavioural strategy for the long term future survival of her genes. Hamilton's first gift to biology was the precise mathematics that a truly Darwinian individual such as a lion would, in effect, have to employ, when taking decisions calculated to maximize the long term survival of its genes. In this book I used informal verbal equivalents of such calculations—on the two levels.

On page 168 we switch rapidly from one level to the other:

> We have considered the conditions under which it would actually pay a mother to let a runt die. We might suppose intuitively that the runt himself should go on struggling to the last, but the theory does not necessarily preict this. As soon as a runt becomes so small and weak that his expectation of life is reduced to the point where benefit to him due to parental investment is less than half the benefit that the same investment could potentially confer on the other babies, the runt should die gracefully and willingly. He can benefit his genes most by doing so.

That is all individual-level introspection. The assumption is not that the runt chooses what gives him pleasure, or what feels good. Rather, individuals in a Darwinian world are assumed to be making an *as-if* calculation of what would be best for their genes. This particular paragraph goes on to make it explicit by a quick change to gene-level personification:

> That is to say, a gene that gives the instruction 'Body, if you are very much smaller than your litter-mates, give up the struggle and die' could be successful in the gene pool, because it has a 50 per cent chance of being in the body of each brother and sister saved, and its chances of surviving in the body of the runt are very small anyway.

And then the paragraph immediately switches back to the introspective runt:

> There should be a point of no return in the career of a runt. Before he reaches this point he should go on struggling. As soon as he reaches it he should give up and preferably let himself be eaten by his litter-mates or his parents.